

Air Traffic Control System Cyber Security Using Humans and Machine Learning

Garrett Atkins
Dept. of Cyber Intelligence and Security
Embry-Riddle Aeronautical University
Prescott, AZ, USA
garrett.atkins@my.erau.edu

Krishna Sampigethaya
Dept. of Cyber Intelligence and Security
Embry-Riddle Aeronautical University
Prescott, AZ, USA
sampiger@erau.edu

Abstract— Aviation performance depends on education and training of pilots and air traffic controllers (ATCOs) on technical, procedural, crisis management, decision making, leadership and communication skills. Recent incidents and studies, however, show crew members may be caught unaware and errors in human judgement can still occur in air traffic control (ATC) systems even more so in the presence of cyberattacks. This paper focuses on raising situation awareness and decision making of ATCOs with cyberattacks impacting operations in next-generation ATC systems. Our goal is to investigate crew-based cyber security as a layer in making ATC systems resilient to cyber and cyber-physical attacks. Automatic Dependent Surveillance Broadcast (ADS-B) technology is used as the basis of our preliminary investigation due to its widely known security concerns and central role in ATC systems. We present experimental study considerations for assessing cyber readiness and improving training of ATCOs. We propose a Kalman filter based cyber alarm solution approach for a machine learning based aid for ATCOs towards early cyberattack detection and incident response in ATC systems.

Keywords—ADS-B, air traffic control, crew, cyber security, machine learning

I. INTRODUCTION

For decades, pilots and air traffic controllers (ATCOs) have been trained to move aircraft safely, securely, and efficiently in airspace, using radar and voice, in the presence of anomalies from nature and accidental system failures. Next-generation Air Traffic Control (ATC) systems, however, require crew skilled to operate in cyberspace with big data, automation, and autonomy (see Fig. 1) [1]. Crew must trust information exchanges on new CNS-ATM technologies—such as Controller Pilot Data Link Communications, Global Navigation Satellite System (GNSS), and Automatic Dependent Surveillance Broadcast (ADS-B)—and interact with a complex network of computing and software for awareness and decision making [1].

A clear and present danger in these new cyber-enabled ATC systems is cyberattacks that elevate existing residual risks and raise new ones. Cyber vulnerabilities and a continuously evolving threat landscape put this pilot and ATCO reliance on systems for critical job tasks at peril, inevitably causing new safety and security concerns. Examples are recent incidents involving accidentally corrupted avionics data [2] and studies about attack impacts of crew's false airspace perception [3].

This work was supported by the U.S. Department of Defense [H98230-20-1-0393 and H98230-21-1-0239].

XXX-X-XXXX-XXXX-X/XX/\$XX.00 ©20XX IEEE

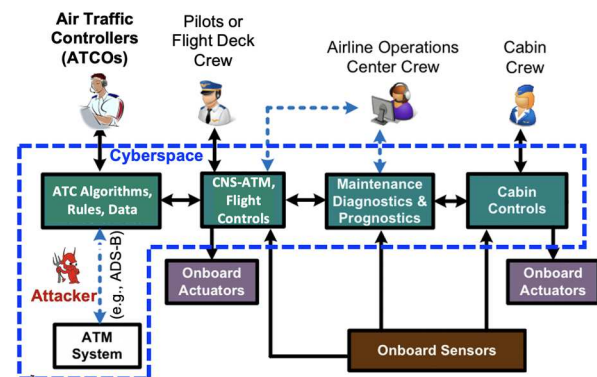


Figure 1. Aviation crew interacting with cyberspace.

Humans as cyberspace users are considered a weak link exploited by cyberattacks, e.g., 95% of cybersecurity issues are traceable to human error (2022 World Economic Forum Global Risk Report [4]). Information Technology (IT) security training of users attempts to prevent users from becoming a victim to well-known attack vectors, e.g., phishing and malware.

However, for an Operational Technology (OT) system, such as aircraft and ATC systems, user training should also consider cyber-physical attacks [1]—a class of cyberattacks that target or leverage the physical domain—e.g., GNSS spoofing and jamming [1]. Training OT crew to identify common attack vectors and recognize signs of cyberattacks can help respond early to incidents before they cause irrecoverable damage to the OT system controlled physical system, environment, and safety.

Humans on their own, however, cannot easily detect threats in cyberspace due to big data and the complexity involved. Automated methods, such as Machine Learning (ML) based approaches, are effective for this purpose, quickly identifying vulnerabilities and attacks likely missed by other means [5].

In this paper, hence, we investigate how ATCOs and ML can together be cyber defense layers that make future ATC systems resilient to malicious attacks from cyberspace. Training ATCOs to be cyber-ready can help detect threats early and allow the system to withstand, recover from, and adapt to adversities and compromises. On the other hand, efficient ML algorithms, working to quickly detect known attacks hidden in big data and flag potential threats, can aid crew in critical ATC scenarios.

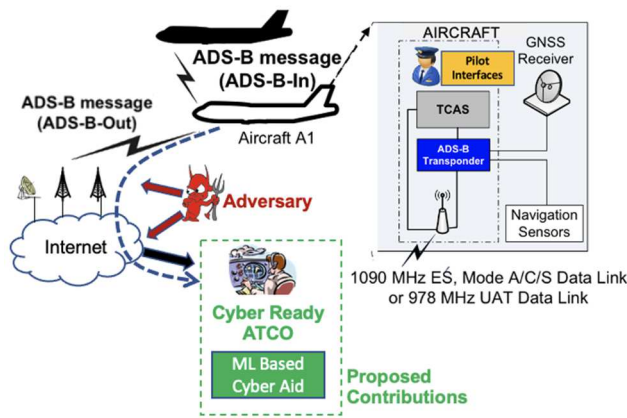


Figure 2. Proposed Approach to Secure ADS-B.

A. Using Crew and ML to Secure ADS-B Applications

ADS-B replaces or boosts radar-based ground and airborne surveillance. GNSS-capable aircraft periodically broadcast their position information (ADS-B-Out) to receivers (ADS-B-In), enabling aircraft tracking at unprecedented resolutions, distances, and airspace regions (see Fig. 2). Due to its intended broadcast use in surveillance applications ADS-B messages are plaintext, i.e., no message authentication or encryption. An attacker, hence, can inject malicious ADS-B messages either over radio or via cyberspace, and threaten to make false air traffic appear on crew displays and as inputs to algorithms.

The fusion and cross-verification from multiple data sources, such as multilateration stations, provide a mitigation for ADS-B ground surveillance applications. However, securing this ground surveillance against Internet intrusions and ADS-B airborne surveillance over radio remain as open problems [1]. We aim to bridge these gaps using ADS-B threat mitigation layers including cyber-ready crew and ML based crew cyber aids.

B. Paper Contributions

- A holistic, resilient approach to ATC system cyber security where ATCO and ML can help combat cyber and more nefarious cyber-physical attacks.
- Insights into an experimental framework to raise ATCO awareness (e.g., how cyber threats impact their situation assessment and what course of action choices they have) as well as to practically assess crew cyber readiness in representative operational scenarios.
- Investigation of a machine learning, i.e., Kalman filter, based approach as an elemental cyber aid for ATCOs to quickly detect potential ADS-B anomalies.

C. Paper Outline

Section II provides some background on related CNS-ATM cyber security concerns. Section III presents challenges and opportunities of crew-based ATC cyber security. Section IV presents the ATC system and threat model. Section V discusses preliminary work on our experimental platform for assessing ATCOs cyber readiness. Section VI and VII present our ML approach to combating high risk threats and aiding crew. Section VIII includes a discussion of challenges, opportunities, and research directions, before the conclusions in Section IX.

II. CURRENT STATE OF ATC SYSTEM CYBER SECURITY

A. ATC System Cyber Security Concerns

The air traffic density is expected to increase exponentially with new advanced airframes taking flight and making air traffic management more complex than ever. The FAA is working to modernize its ATC/ATM systems through the next generation (NexGen) air transport system. This modernization effort will require the replacement of current isolated systems with Internet based information systems to monitor and control aircraft.

Cyber threats to aviation have been on the rise for over a decade. While past articles, such as [6]-[9], raised alarm to the vulnerability of current ATC systems, it was argued these ATC systems are relatively isolated making them less discoverable by remote attackers. For the NextGen transition, the U.S. Government Accountability Office (GAO) continues to examine FAA's cybersecurity efforts and provides recommendations, such as [10], [11]. Areas of concern are the cybersecurity of new Internet connected ATC systems in the NextGen as well as related aircraft systems such as ADS-B and GPS.

GAO highlighted cybersecurity challenges in multiple areas, including protecting the ATC information systems and services. Information systems using the Internet and its underlying protocols and technologies are well known to be vulnerable to a continuously expanding and evolving cyber threat landscape.

ADS-B has long been known to be vulnerable [12], not only due to its reliance on a vulnerable GPS, but also its inherent broadcast nature that allows cyberattacks such as spoofing, jamming, flooding, and injection. Furthermore, cyber security of data link communications, such as the CPDLC, as well as ground-based navigation aids used by aircraft are concerns, including Instrument Landing System (ILS) [13] and modern Ground-Based Augmentation System (GBAS) [14]. These systems have experienced failures from accidental causes, but their vulnerability to intentional attacks remains unclear.

Currently, there seem to be no publicly accessible plans for making existing and the NextGen ATC systems resilient to cyber threats. The FAA's Air Traffic Organization (ATO) Cybersecurity Group (ACG) is charged with establishing ATO security services but it remains unclear how ATC systems will be secured [15]. In addition, the FAA and the MITRE Corporation detailed a cyber architecture regarding the security of National Airspace System (NAS), however, it does not mention architecture to protect ATC systems specifically [16]. Unfortunately, recent incidents show ATC systems may not be robust against accidental data corruption and signal jamming.

B. Accidental Cyber Disruptions in ATC Systems

In January 2023, FAA's Notice to Air Missions (NOTAM) system experienced an outage resulting in the grounding of aircraft in the National Airspace System (NAS) [17]. While evidence of a cyberattack was not found, the FAA noted the outage was caused by an accidental corruption of a database file [18]—a common effect shared with those from cyberattacks. This incident exposed the dependence of ATC system continuity and performance on cyber accuracy and functionality as well as potential cyberattack impacts such as the cascading effect of cyber disruptions in the NAS.

Recent incidents also show how vulnerabilities in new CNS-ATM technology can disrupt the ATC systems. Examples include GPS jamming at the Denver International Airport for 33 hours [19] and Dallas-Fort Worth (DFW) International Airport for 44 hours [20]. At Denver, pilots and ATCOs reported GPS, Traffic advisory and Collision Avoidance System (TCAS), and other system disruptions. FAA sent NOTAMs advising about disruptions such as GPS interference up to 40,000 feet above mean sea level around Denver as well as ADS-B unreliability. In the GPS interference incident at the DFW airport, flights were reported to have been rerouted and, unlike Denver, the jamming source remains publicly unknown at this time [20].

C. Crew Related Concerns for ATC Cyber Security

Crew play an active role in managing aircraft flight and air traffic, with pilots and ATCOs using CNS-ATM technologies, such as radar, multilateration, ADS-B, and voice and data communications to coordinate with each other and airline employees. These crew members assuredly help manage safety and operational risks from nature and humanmade physical threats. Recent incidents, however, reveal this interaction has residual risks greater than expected. Pilot and ATCO involved incidents [21], emerging crew concerns, such as aging, fatigue and workload [22], and automation addiction [23], put into debate the reliance on crew reacting to abnormal events.

Crew rely on their own experience and capability as well as cues and other information from their systems to observe, assess situations, and choose a course of action to assure systems are functioning nominally. However, to the best of our knowledge, pilots and ATCOs have no cues or solutions in their interfaces to detect anomalies from potential CNS-ATM cyberattacks and also lack the proper training to detect signs of cyber intrusions and respond to cyberattack impacts in operational environments.

III. TOWARDS CREW-BASED ATC CYBER SECURITY

A resilient approach to ATC system cyber security requires awareness, education, and training of crew who are inevitably users of systems under cyber and cyber-physical attacks. A major aviation industry opinion is that cyber security tasks for crew in high workload and stress professions, including pilots and ATCOs, causes unnecessary work. These crew members are regularly given knowledge and skills to operate, maintain, and manage, multi-million-dollar assets, yet they mostly lack the basic knowledge and fundamentals of cyber defense. These crew members, hence, require a holistic training to optimally perform and execute highly specialized skillsets, intellectually, physically, and psychologically, in the presence of cyberattacks.

A general approach taken by standards is to use existing advisory systems to notify pilots about system failures. Today these failures are from accidental causes, but cyber resiliency requires consideration of system failures intentionally caused by cyberattacks. Pilot advisories are given only if a failure requires an immediate or a deferred action from a pilot during flight. But no insight into the “under the hood” cause of this failure is given to pilots. Only ground crew, e.g., for maintenance, access the impacted systems when the aircraft is on the ground to gain such insights. Moreover, an important distinction between accidental and malicious causes are the unreliability of the presented advisories and unpredictability of the outcomes of failure.

Not surprisingly, pilots have begun to express the need to be aware and in control of cyber security incidents impacting their flights [24]. They want to be trained to identify and handle cyber security incidents. Recent studies warrant that we seriously consider enabling them. Some pilots were observed unable to detect disruptions caused by cyberattacks in a flight simulator environment, such as when navigation database and GNSS receiver data were spoofed [3], [25]. Making pilots aware of ongoing cyber security incidents can be beneficial and reduce pilot workload, compared to not informing them.

Recent articles indicate an initial interest in the industry to develop tools to train pilots to respond to aircraft system cyberattacks [26]. An onboard intrusion detection system that warns pilots about aircraft system hacking attempts and malicious behaviors is also being investigated [27].

On the other hand, a recent study shows ATCOs too may fall victim to cyberattacks in operative scenarios and unaware of how the situation evolves [28]. Cyber incidents, illustrated in a desktop exercise using surveillance pictures, negatively impacted ATCO workload, situational awareness, and trust in ATC systems. Raising crew awareness and adding cyber security considerations in ATCO training were recommended.

Unlike with pilots, however, there are no advances towards introducing ATCO cyber security tasks, training, or alarm tools. The FAA’s Air Traffic Collegiate Training Initiative (AT-CTI) program, which trains future ATCOs, lacks security content in its curriculum [29]. Furthermore, the NextGen ATCO training (classroom, online, simulator, and on-the-job) plans and investigations do not include cyber security considerations [30].

Hence, we address this gap and propose a crew-based cyber security approach. But, in doing so, it is important to balance considerations between cyber-physical security, autonomy, and crew. Cyber-physical threats target system state, behavior, and performance, e.g., malicious data in avionics and ghost aircraft in airspace, hence impacting autonomy and crew cognizance. Whereas, the design of autonomous threat monitoring and response, e.g., rapid detection of cyber intrusions in aircraft and ATC, can aid crew to manage contingency events and situations from cyber-physical attacks. Recent aviation developments also warrant that it is crucial to integrate crew with autonomy and cyber-physical security for safety and resilience [2], [3], [25].

This paper, hence, takes steps towards achieving this balance and understanding the complex symbiosis between cyber-physical security, autonomy, and crew in ATC systems. A critical step is to embed cyberattack scenarios in ATCO training that can help them understand how to detect and respond to cyberattacks. Another critical step is that cyber situational awareness via a cyber aid be introduced in an effective, but non-intrusive, manner into the ATC operational environment.

We investigate using some existing ATC simulators, capable of simulating cyberattacks, as an educational platform to raise awareness about ATC cyber security and as an experimental platform for ATCO cyber readiness assessment and training. These platforms expose users to malicious data during normal ATC operative scenarios. The resulting experimentation can provide insight into the observed threat impact on ATCO situational awareness, decision making, and workload.

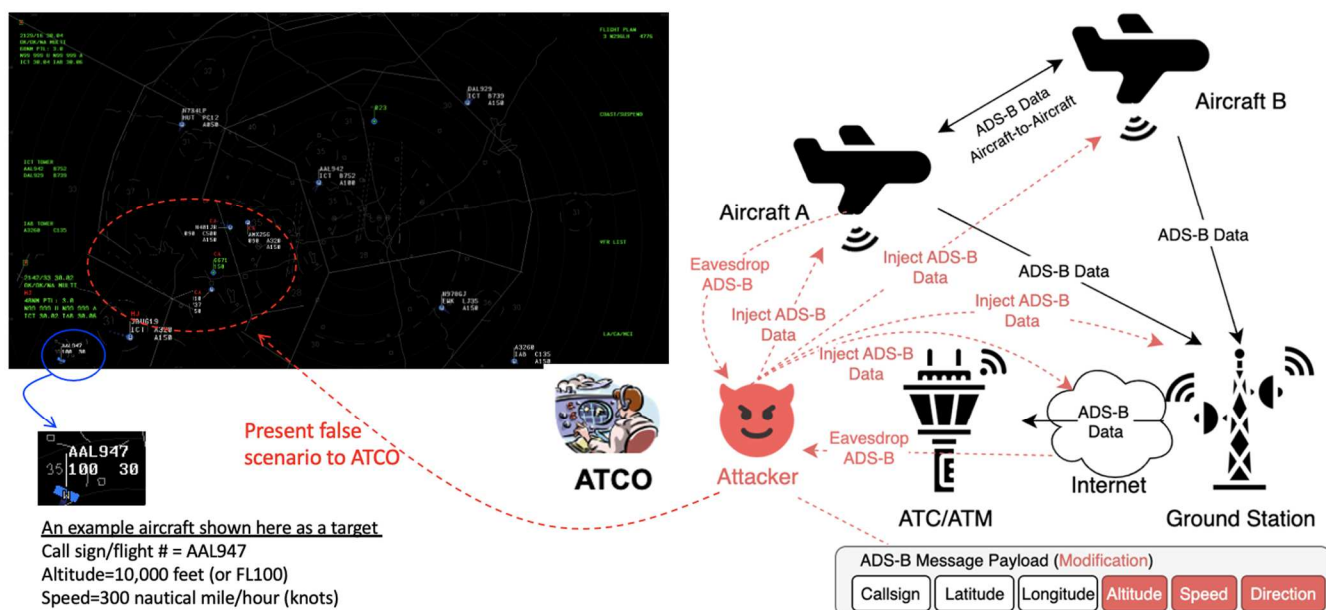


Figure 3. Illustration of threat scenario considered. Attacker is eavesdropping and injecting ADS-B messages over radio and the Internet with the goal of creating false air traffic conditions to the ATCO.

IV. ATC SYSTEM AND CYBER THREATS CONSIDERED

A. System Model Considered

ADS-B enabled aircraft exchange their current position information automatically and openly with airborne and ground receivers. A set of geographically spread ADS-B ground stations (e.g., over 700 across continental USA) collect and send data through a telecommunication network, using Internet protocols, IP packets, and technologies, on to servers for further processing and ultimately to ATC displays.

Within each ATCO controlled airspace sector and area, specific altitudes, speeds and vectors are allotted for arrival and departure aircraft. An ATCO may be responsible for a subset of aircraft, such as aircraft departing, arriving, on in final approach. The ATCO may interact with other controllers for tasks such as transfer control of arrivals to the Final Approach controller. The ATCO may be responsible for multiple airspaces. Furthermore, ATCOs may perform additional coordination between themselves, such as ensuring crossing restrictions and airspace limitations are met. ATCOs coordinate with pilots for instructions, clarifications, and updates.

Figure 3 (left side) shows an example display where an ATCO would see aircraft in flight. Each aircraft appears as a target displaying information, such as call sign (e.g., AA947), type (e.g., B737), and status (e.g., squawk code, speed, heading, and altitude). Their flight plans are displayed for ATCO access and priority. A flight plan contains information about the aircraft, its pilot, and its flight such as the planned departure, destination, estimate cruise time, planned route, altitude, and speed (https://www.faa.gov/air_traffic/publications/atpubs/). Trailing dots seen in the screen are aircraft position tracks captured by primary radar or reported over ADS-B. Trailing dots are relatively closer, compared to radar, since ADS-B position information is broadcast by each aircraft once every second.

Using these types of display ATCOs are educated and trained to monitor and recognize air traffic conditions, perform ATC tasks—such as sequence departures onto requested transitions, climb to either the requested altitude or the airspace ceiling, and maintain separation from other aircraft and airspace sectors—and interact with each other and with pilots.

B. Threat Model Considered

ADS-B is vulnerable and lacks security primitives including message encryption and authentication, and entity authentication. These weaknesses together with the vulnerable ground-based network and information technology, enable an attacker to attempt to spoof, jam, flood, and inject false ADS-B messages. A malicious actor, with an extensive knowledge of ADS-B protocol and message format, can directly attack the radio communication or modify ADS-B messages on network.

We assume our adversary's primary goal is to make ATCOs perceive false ADS-B air traffic conditions as genuine traffic. The adversary wants to cause inaccurate actions by ATCOs, interrupt ATC operations, threaten physical destruction (e.g., near-misses), and force aircraft ground stops.

Towards this goal the adversary may attempt to either inject one or more ghost aircraft, i.e., aircraft that do not exist physically, jam aircraft (either at the physical layer or higher layers of the Internet stack), or corrupt information of physical aircraft on the screen. ATCO will not receive any response from the pilots of ghost aircraft since there are no real aircraft.

We assume the adversary conducts a reconnaissance phase where the ADS-B data of aircraft flights is passively intercepted, in air or over ether, and recorded. Then, in an active exploitation phase, this recorded data is tampered and replayed with the intent of making them appear on the ATCO display.

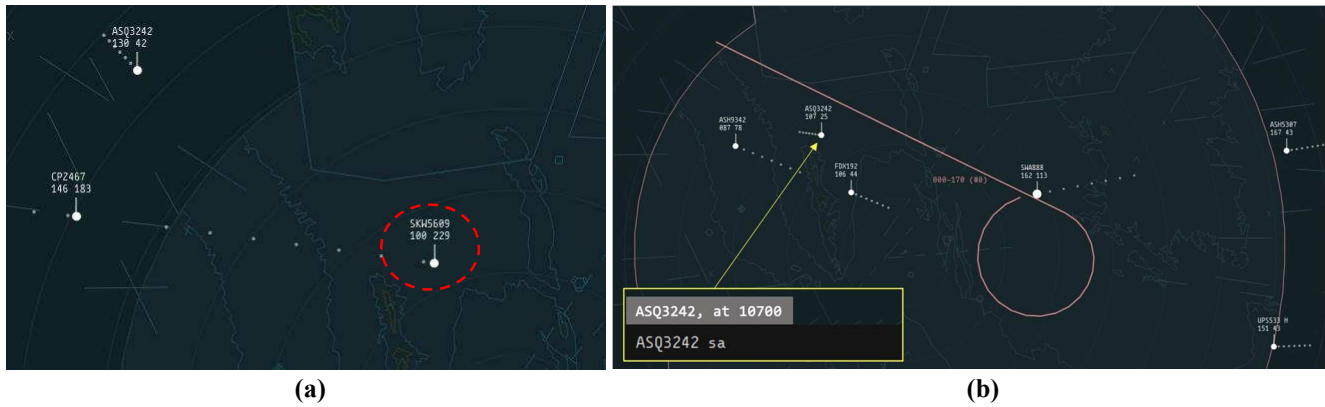


Figure 4. Raising ATCO cyber awareness. (a) OpenScope simulator shows moving aircraft with tailing dots. Each aircraft label shows its call sign, speed (in tens of knots), and altitude in hundreds of feet. A ghost aircraft, i.e., SKW5609, is identified due to its unusual mobility and much faster than normal indicated by tailing dots. (b) Ghost aircraft shows unusual mobility and does not respond to ATCO communication, only genuine aircraft (e.g., ASQ3242) respond.

1) *Reconnaissance: Eavesdropping*: Figure 3 (right side) shows the attack scenario of a malicious actor performing a eavesdropping attack against ADS-B communications. Given the broadcast nature of ADS-B, an attacker can easily and passively intercept and collect ADS-B transmissions to track aircraft. Furthermore, ADS-B flights are tracked and made accessible by several web servers on the Internet (e.g., FlightRadar24). Eavesdropping can be considered harmless due to its passive nature, but it can be utilized by an attacker to gather intelligence and launch more sophisticated attacks.

For example, an attacker can eavesdrop on an ADS-B frequency to gather information about all the real-time air traffic of a specific geographical area, or can perform reconnaissance on a particular aircraft. An attacker can then use this gathered information to launch a cyber (e.g., flight privacy violation), physical (e.g., surface-to-air weapon use), or cyber-physical attack (e.g., spoofing of air traffic) against aircraft or ATC systems, e.g., replay the recorded flight to appear on the ATCO display on a different day.

2) *Exploitation: Spoofing via Message Injection*: Figure 3 (right side) shows an attacker who crafted an ADS-B message by modifying only the altitude, speed, and direction in the ADS-B message's payload and injecting these fake messages into the system as part of the exploitation phase of the attack. The lack of ADS-B message authentication and entity authentication makes this type of spoofing a threat. An attacker with the knowledge of the ADS-B protocol and its system architecture can, for instance, inject these malicious messages into the system either via radio or through the Internet and increase risk.

V. TOWARDS ASSESSING ATCOS CYBER READINESS

We propose to develop and incorporate the just described threat scenarios into an ATC simulator to: (1) raise awareness of ATC cyber threats and (2) assess ATCO cyber readiness. Our ultimate goal is to conduct experiments where we can observe and evaluate how ATCOs react and behave to them, both with and without alarms through cyber aids and prescribed responses; but this will be conducted as a future work.

With experimentation, we expect to: (1) gain insights on how cyberattacks impact crew, such as in terms of situation

awareness, workload, behavior, attention, stress, and trust; (2) assess crew's capacity to learn and understand cyberattacks, analyze cues and choose response options, and effectively manage their response to stressors from cyberattacks; (3) ensures the training exercises enables crew to make informed choices that directly influence and enhance their skillsets, productivity, and job performance in the presence of cyberattacks; and, (4) design cyberattack mitigations that carefully integrates the role of crew.

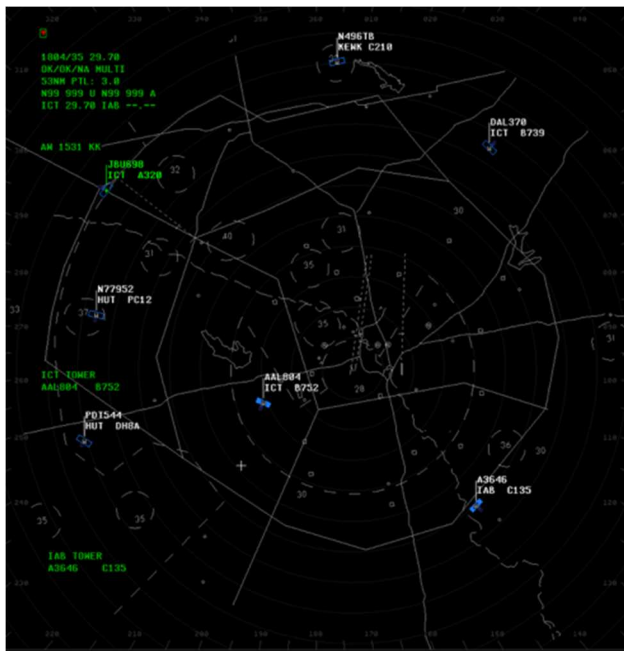
Cyber security considerations in training exercises on a simulator setup, for instance, can include adding an unexpected ATC system malfunction or a failure cue and informing the ATCO that a cyberattack is suspected. Such a scenario helps raise cyber awareness and train students to manage ATC operations under the resulting unpredictable, false alarm, and low trust conditions. For more invasive, but covert, cyberattacks, such as those discussed earlier about displaying spoofed data on screen, we plan to use various threat scenarios in departures, arrivals, final approach, control transfer, and cruise, to qualitatively assess crew reactions and responses.

A. Experimental Platform for ATC Cyber Threat Awareness

OpenScope is an experimental ATC platform that can be used to simulate ADS-B based cyberattacks [31], [32]. It helps specifically to simulate spoofing attacks, where ghost ADS-B nodes appear on a screen containing cooperative aircraft departing and arriving at a given airport. Scenarios can be offered in this context to ATCOs and their reactions and responses can be evaluated in the presence of ADS-B spoofing.

Figure 4 (left side) shows a scenario where a ghost aircraft is speeding through the ATCO display presenting status information. Trained ATCOs are expected to recognize such abnormal aircraft and air traffic conditions. For instance, the ghost aircraft is clearly defying physics in its speed. Other conditions might be that it is not presenting valid flight information or does not have a filed flight plan.

Figure 4 (right side) shows how the ghost aircraft continues to demonstrate unusual flight patterns. Furthermore, the ATCO would also not be able to communicate with the ghost aircraft, but only with genuine aircraft such as the one shown.



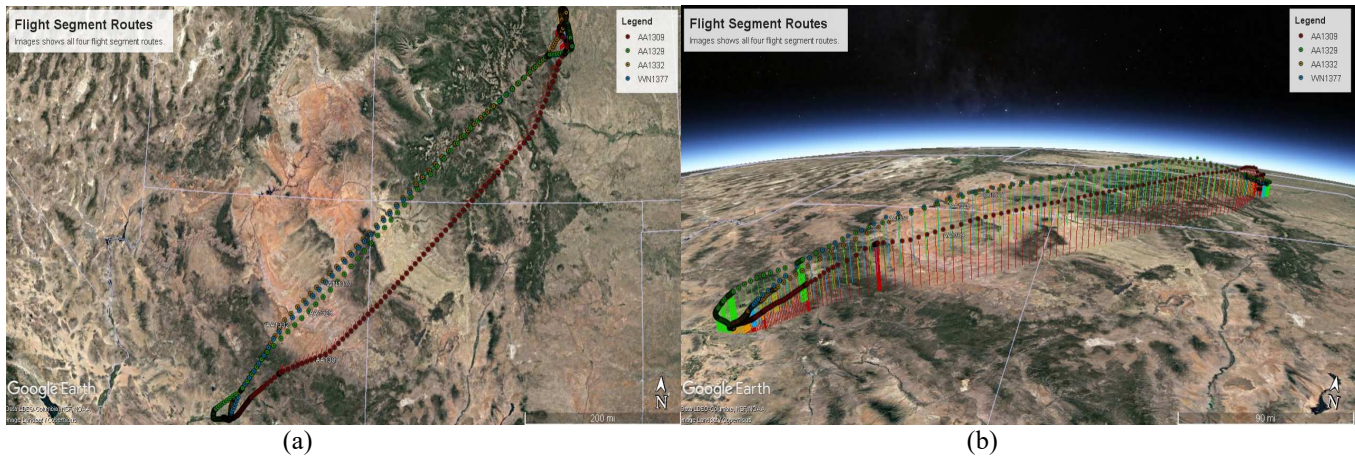


Figure 6. Flight Segment Routes Mapped in 2-D (a) and 3-D (b).

The development of a system to detect cyber intrusions would be beneficial and reduce the chance of a disaster, compared to humans using observation and intuition to detect them. Protecting ATC systems from cyberattacks, however, is a complex problem with a solution that has yet to be realized and implemented. It is vital that the detection system does not interfere with the normal flow of ATC operations and ATCO workload. Furthermore, a cyber aid for ATCOs must be effective at providing enhanced cyber situational awareness while providing relevant information for incident response.

We explore preliminary use of a machine learning model to detect spoofed ADS-B data, using simulated and real data to train and test the considered model. The machine learning based intrusion detection system will be evaluated by testing various attack scenarios which manipulate ADS-B message data. This demonstration can provide valuable knowledge about whether such a system is advantageous to detect an ATC cyber attack but also capable of assisting ATCOs in identifying one.

The two attack scenarios we focused on are eavesdropping and spoofing via message injections described in Section III. We exercise the use of Kalman Filters to detect ADS-B spoofing attacks on the ground-based ATC system. It is important to note that our implementation does not address message privacy to prevent eavesdropping attempts as it requires modification to ADS-B technology architecture. However, we consider eavesdropping as an attack scenario because it can be used by an attacker to perform a spoofing or replay attack. We aim to demonstrate the practicality of a ML-based cyber aid to detect anomalous ADS-B messages and provide trustworthiness thresholds to enable cyber situational awareness in ATCOs.

VII. PROPOSED KALMAN FILTER BASED ATTACK DETECTION

As Kalman filter predictions are based on a physical dynamic model of the past data, the prediction error of a Kalman filter can be employed to detect attacks. We exploit this concept to show how Kalman filter based detectors can be employed as an algorithm for detection of spoofing or replay attacks in ADS-B data.

1) *Dataset Selection:* Flightradar24 was the tool selected to provide the authentic messages dataset. Flightradar24 is one of the largest global ADS-B networks which provides real-time

flight tracking information about hundreds of thousands of aircraft per day to its users. This service allows a user to subscribe to the platform in order to export historical and current ADS-B data for aircraft flight segments in .csv or .kml format. The .csv format contains the following ADS-B parameters: Timestamp, callsign, latitude, longitude, altitude, speed, and heading.

We used the Flightradar24 service to identify flight segments where the aircraft's origin was Phoenix Sky Harbor (PHX) and its destination was Denver International (DEN). A short haul flight segment was chosen to limit the amount of data so that general flight phases are seen in a shorter period of time compared to a long haul flight. Further, Kalman Filters do not need a vast dataset(s) compared to other learning algorithms. We selected four different PHX to DEN commercial flight segments to be used as the authentic messages' dataset. Figure 6 illustrates the aircraft routes mapped inside Google Earth Pro and Figure 7 shows the authentic message dataset format.

Timestamp UTC	Callsign	Latitude	Longitude	Altitude	Speed	Direction
1677120118 2023-02-23T02:41:58Z	AAL1309	39.83382	-104.759033	7625	198	244
1677120121 2023-02-23T02:42:01Z	AAL1309	39.832748	-104.761902	7675	200	243
1677120124 2023-02-23T02:42:04Z	AAL1309	39.83139	-104.765388	7750	201	243
1677120127 2023-02-23T02:42:07Z	AAL1309	39.830063	-104.768791	7825	203	242
1677120130 2023-02-23T02:42:10Z	AAL1309	39.82893	-104.771606	7875	204	242

Figure 7. Authentic Messages Dataset Format

We chose datasets that reflect a complete commercial flight segment which includes all phases of flight from takeoff to taxi. The previous studies and research mentioned in the problem description section only focus on the enroute or cruise phase of the flight. We believe focusing on the entirety of an aircraft's flight is more appropriate because an attacker can launch an attack on ADS-B at any time, not just when aircraft is in cruise.

For the attack messages dataset, a Python script was developed to produce modified ADS-B messages and inject them into the authentic messages' dataset. Figure 8 shows the design and functionality of the Python script to modify and insert falsified messages into the input dataset. Figure 8 also shows the output of attack dataset generation script.

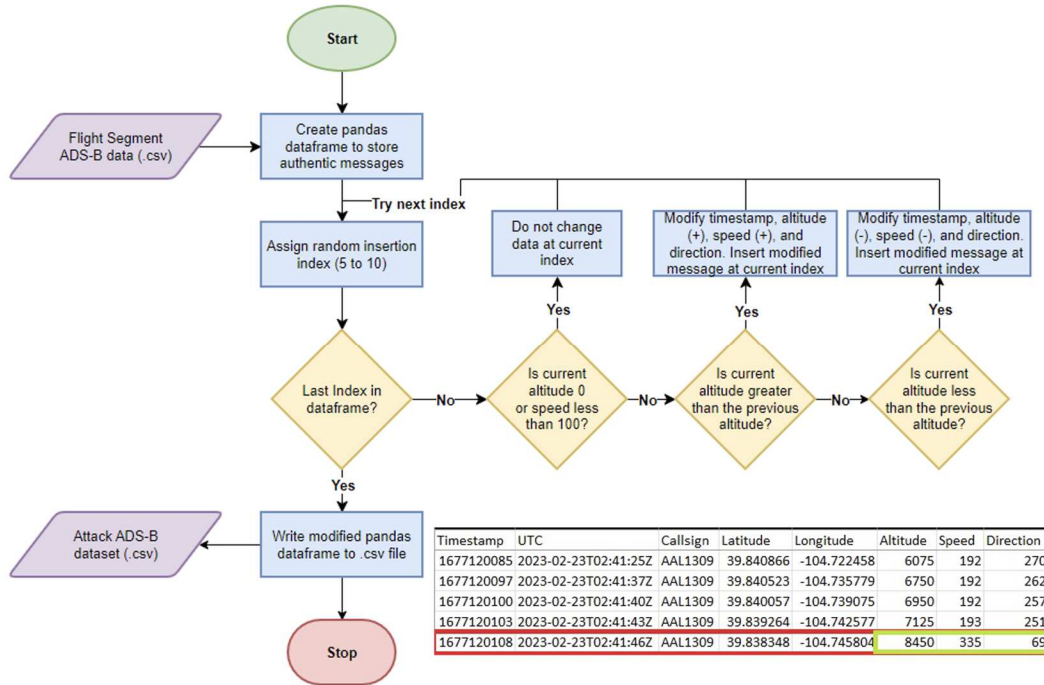


Figure 8. Attack Dataset Generation Script Flowchart and Example Output with an Offset Value of 5.

The three parameters of the ADS-B dataset which were focused on were altitude, speed, and direction. These parameters were selected to simulate our spoofing attack via message injection. The script was written as if we were an intermediate attacker who understands the ADS-B message format and is skilled enough to randomize the insertion and selected parameter values. The script selects a random insertion value between five through ten (5-10) every iteration and based upon the insertion value it selects a random value from a list of provided offsets for altitude and speed. The direction parameter is modified by changing the value to the opposite. We used the authentic messages datasets inputted into the Python attack generation script to generate four unique attack datasets. Table 1 provides detail about the selected parameters offsets.

2) *Kalman Filter Model:* Current cross-verification methods, such as multilateration, heavily rely on radio signals which is not effective at verifying digital signals. On the other hand, the use of a sophisticated model which considers aircraft dynamics is known to be computationally inefficient. Further, ADS-B Out broadcasts aircraft location data once per second to ground stations and neighboring aircraft. It is vitally important that the ADS-B dataflow is validated in a timely manner so ATCOs can issue accurate traffic commands to aircraft pilots. We utilize simple dynamic models in our proposed solution. The issue of oversimplification can arise from applying a less complex model to a complex problem, but we aim to mitigate this issue by comparing expected results to actual results.

Aircraft position prediction can be considered as a state estimation problem, hence, the need for an algorithm which can construct a reasonable predicted state from noisy measurement data. If we consider a linear state space model, the Kalman Filter (KF) is an optimal filter. The process and measurement equations are as follows for the linear motion state space model:

$$x(k+1) = A(k)x(k) + G(k)w(k) \quad (1)$$

$$z(k) = H(k)x(k) + v(k) \quad (2)$$

The first equation, the current state $x(k)$ is transformed to the next state $x(k+1)$ by the state transition matrix A and corrupted by process noise gain matrix $G(k)$ multiplied by the process noise $w(k)$. The second equation is the measurement equation which describes the system output $z(k)$ is related to internal states through the measurement matrix H and measurement noise $v(k)$. In the initialization of the Kalman Filter, we assume the process noise and measurement noise are Gaussian.

The Kalman Filter is a two-stage recursive algorithm. At each iteration or step, the algorithm performs a state prediction followed by a state correction. The algorithm also computes and updates an important element known as the Kalman Gain which helps minimize estimate variance.

Table 1. Attack Dataset Parameter Offset Values

Altitude Offsets (feet)	Speed Offsets (GS in Knots)	Direction Offset (heading in degrees)
200	20	(Direction + 180) % 360
400	40	
600	60	
800	80	
1000	100	
1200	120	
1400	140	
1600	160	

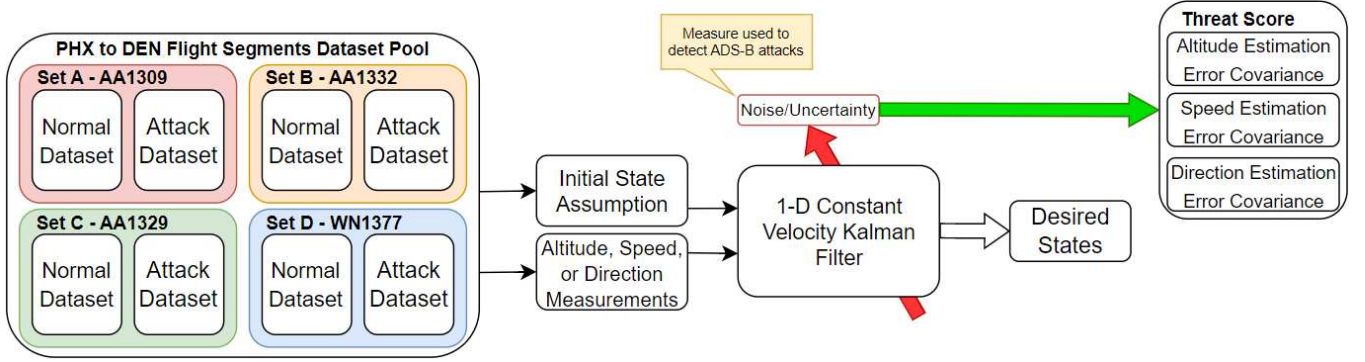


Figure 9. Diagram of Dataset Pool with High-level Diagram of Kalman Filter

First, the algorithm requires the initial system state $x_{0|0}$ and initial state uncertainty $P_{0|0}$ where $x_{0|0}$ is set to the first indexed ADS-B message parameter (altitude, speed, or direction) in the dataset while $P_{0|0}$ is set with positive values in the matrix diagonals.

The prediction stage:

$$x_{k+1|k} = F_k x_{k|k} + B_k u_k \quad (3)$$

$$P_{k+1|k} = F_k P_{k|k} F_k^T + G_k Q_k G_k^T \quad (4)$$

$$z_{k+1|k} = H_{k+1} x_{k+1|k} \quad (5)$$

And the correction phase:

$$S_{k+1} = H_{k+1} P_{k+1|k} H_{k+1}^T + R_{k+1} \quad (6)$$

$$K_{k+1} = P_{k+1|k} H_{k+1}^T S_{k+1}^{-1} \quad (7)$$

$$x_{k+1|k+1} = P_{k+1|k} + K_{k+1} (z_{k+1} - z_{k+1|k}) \quad (8)$$

$$P_{k+1|k+1} = P_{k+1|k} - K_{k+1} S_{k+1} K_{k+1}^T \quad (9)$$

After initialization, the filter generates the state to the next step utilizing the motion equation followed by generating the covariance matrix. The fifth equation is used by the filter to output a predicted measurement z_k value at $k + 1$ step. Next, the filter compares the difference between the actual measurement and the predicted measurement to correct the state at $k + 1$ by computing the estimation error covariance and the Kalman Gain. The filter uses the measurement and corrects the predicted estimate z_{k+1} along with the Kalman Gain K_{k+1} . The filter lastly performs a correction of the state covariance matrix. After initialization, the Kalman Filter repeats this process of prediction and correction until the length of measurements ceases.

Before implementation of the Kalman Filter, we needed to carefully select the appropriate dynamic model for our aviation application. Typically, commercial aircraft submit a flight plan to FAA's Flight Service organization which specifies speeds, altitudes, and headings for the aircrafts route. The movement of an aircraft can be considered as linear motion; thus, a constant velocity model would be best suited for our application.

For our implementation, we chose to utilize three separate 1-Dimensional constant velocity Kalman Filters and applied them to each of the three ADS-B parameters; altitude, speed, and direction. Figure 9 provides a high-level illustration of the Kalman Filter application. The output of the Kalman Filter we were most interested in was the estimation error covariance also known as the prediction error. The estimation error covariance allows us to determine if an ADS-B attack is present which is discussed in the performance evaluation section.

B. Performance Evaluation

We used three distinct methods to evaluate the performance of our proposed approach, specifically, the ability of our model to distinguish between normal and anomalous ADS-B message traffic. The three methods of performance evaluation were broken down into subsections: Datasets, detection evaluation, and thresholding and trustworthiness scale.

1) *Datasets:* As mentioned earlier, we applied our proposed approach to a total of four different PHX to DEN commercial flight segments. The authentic messages dataset consisted of four flight segments obtained from the Flightradar24 online tool. Three of the flight segments were operated by American Airlines (Airbus A320/1) and one flight was operated by Southwest Airlines (Boeing 737 MAX 8).

The four datasets were then used to create four unique attack datasets by using the attack generation Python script to generate and inject anomalous ADS-B messages. The datasets are considered unique because the script selects a pseudo-random insertion point every iteration through the message modification loop.

Figure 10 illustrates the four sets of datasets where each set consists of an authentic (normal) dataset and a generated corresponding anomalous (attack) dataset. We overlaid the flight segments in a graph to investigate the features of the curves which is seen in Figure 10. The authentic flight segments show similar features despite variances caused by weather, airframe type, route, and pilot maneuvering to name a few variables. There are a few areas where sharp curves are exaggerated on the graph which indicates high prediction error by the Kalman Filter. This high prediction error is caused by sudden changes, increase or decrease, in the ADS-B parameter values (speed, direction, and altitude) in which we are analyzing for attacks.

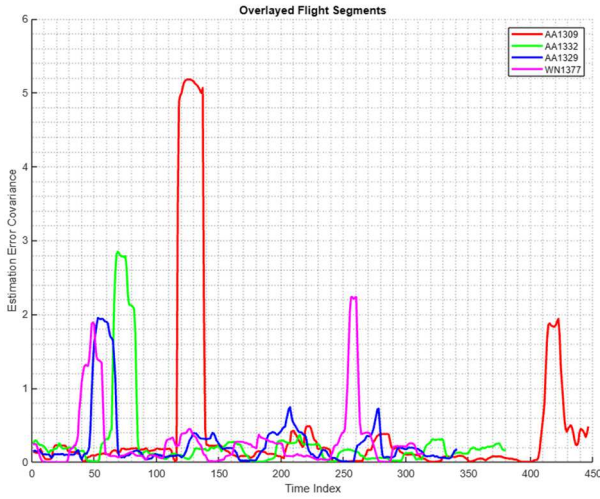


Figure 10. Overlaid segments of the error covariance of Kalman filter in normal flights

1) *Detection Evaluation:* To evaluate the performance of the proposed approach, we focused on the estimation error covariance measures to provide the detection ability. To start, we used the prediction states for the speed, direction, and altitude separately and calculated the standard deviation for each prediction state x_i over a window length N of 20 points.

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}} \quad (10)$$

The standard deviations for speed s , direction h , and altitude z are multiplied together to provide what we are calling a threat score TS .

$$TS = s * h * z \quad (11)$$

The threat score can be considered a measure of threats and is derived from the estimation error covariance calculated by the Kalman Filters for each of the prediction states. Again, the estimation error covariance or prediction error is the amount of deviation from the actual measurements. The threat score is a pseudo-bounded measure in that if the data is bounded the threat score is thus a bounded measurement. However, we have chosen not to bound the threat score as vital detail or features are lost when bounding the measurement.

We experimented with the idea of bounding the threat score measurement $0 \leq f(x) \leq 1$. The following function was used to bound the measurement:

$$f(x) = \frac{2}{1+e^{-x}} - 1 \quad (12)$$

Figure 11 shows the results of using the bounding function on the threat score measurement for the first data set. It can be seen in graph above the amount of detail that is taken away by bounding the measure. The error could be large and one would not be able to see the actual amount of error because of the bounding process. However, the probability of the error being high is low, but the possibility of it being high is reasonable. For these reasons, we decided to keep the threat score measure unbounded so valuable details or features are not lost when the measure was bounded between 0 and 1.

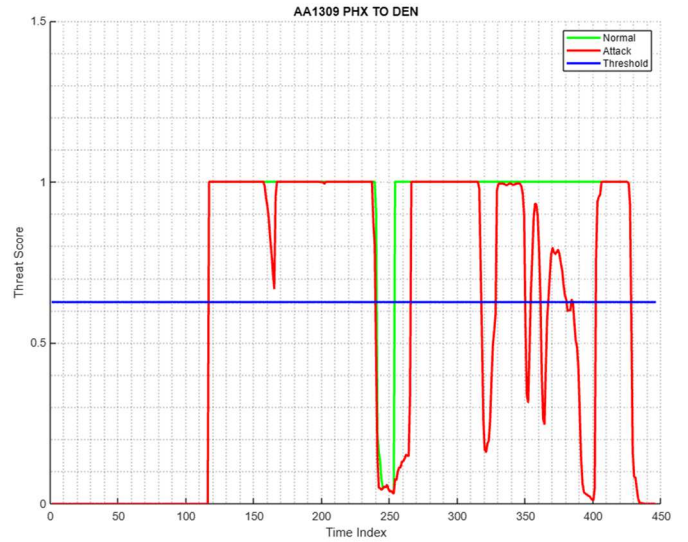


Figure 11. Results of Bounding Threat Score

1) *Thresholding and Trustworthiness Scale:* In a warning or threat detection system, often a threshold level is selected for the threat score to declare the severity of the potential warning or threat. This threshold in turn determines the probability of detection and the probability of false alarm of the detector, or the trustworthiness of the warning system. In order to incorporate the trustworthiness factor of the detector, we can choose multiple (say, 3) levels of thresholds to estimate the level of the threat from measuring the distance of the threat score from the threshold levels, corresponding to high, medium, and low confidence, for instance.

Note the trustworthiness scale is not intended to evaluate the trustworthiness of the model but is intended to provide insight into the results of data. Further, the trustworthiness scale is designed to provide the ATCOs with the ability to make a rapid decision about the accuracy of the information displayed by the ATC systems.

The trustworthiness scale was developed by establishing three thresholds. The thresholds were calculated by taking the mean of the estimation error covariance when the aircraft was in the cruise phase of flight and multiplying it by an arbitrary number n .

$$Threshold = \frac{\text{sum of error covariances}}{\text{number of error covariances}} * n \quad (13)$$

We were able to estimate when the aircraft was in cruise based upon the error covariance measurements were more stable. The mean was taken during the cruise phase of the flight so that the threshold can be more realistic because if the other flight phases were included the threshold would account for the large error when the aircraft is taking off and descending. In addition, an arbitrary number was selected to multiply the mean in order to account for uncontrolled variables which cause an increase in prediction error. Our proposed trustworthiness thresholds and scale for our implementation can be seen in Figure 12.

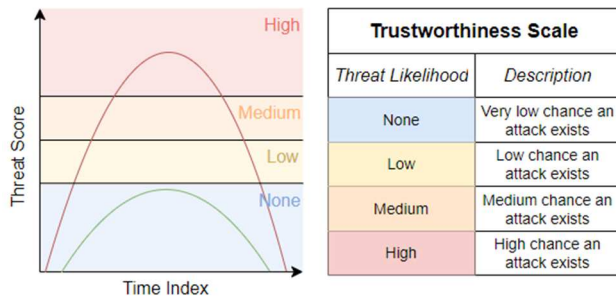
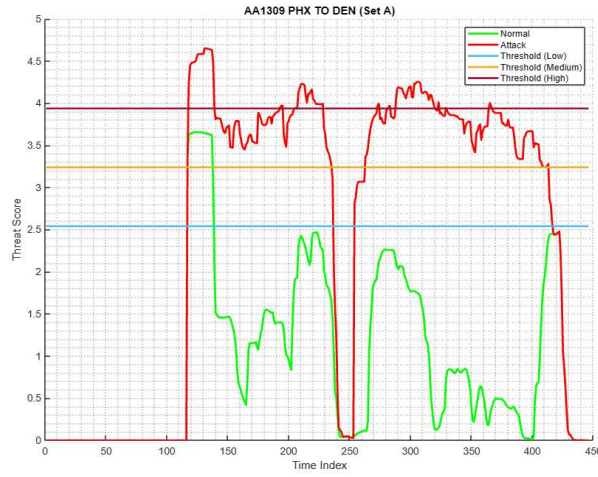


Figure 12. Trustworthiness Thresholds and Scale

2) *Results:* In summary, as shown in Figures 13(a)-(d), our results for each set of data, indicate that the separation between the attack (red) and normal (green) threat score are well separated and the Kalman filter approach works quite well in detecting attacks. It can be seen in the graphs that even though we did not bound the threat score; the score is bounded by the data as the majority of the points for the normal curve fall below a score of 2.5.



(a)



(b)



(c)



(d)

Figure 13. (a) Flight AA1309 Set A results. (b) Flight AA1329 Set B results. (c) Flight AA1329 Set C results. (d) Flight WN1377 Set D results.

The shape oscillations looking at the attack curve indicate an increase in estimation error covariance which represents a possible anomalous ADS-B message. Looking at the attack curve of all four graphs, it is apparent that the message injections were randomized due to the noticeable variability. The calculated thresholds using an arbitrary number of five (5) multiplied with the mean of the cruise measures are appropriate for all datasets in our evaluation.

VIII. DISCUSSION AND FUTURE WORK

A. ATCOs Cyber Readiness: Challenges and Opportunities

Divergence between crew state awareness and actual system state is a known NAS safety risk. A future direction is to explore divergence vulnerabilities when crew is subject to cyber threat scenarios without and with our KL-based cyber aid, with the goal of eliminating divergence causes and hazardous consequences. Measuring crew reactions yields insights, e.g., failures in crew perception, and projection of system state in threat scenarios and mitigations, helping improve its design. We also address potential multi-agent divergence issues, e.g., inconsistencies in state awareness of pilot and ATCOs.

Furthermore, a cyber aid must be easy to use and integrate well into crew duties and procedures. The intelligence added to the ATC display screen must be intuitive in the rendered crew situation awareness about something the crew don't see and easily assist them only when an emerging situation warrants it; e.g., alerts about a physical impact of cyberattacks and procedures to turn off or reset misbehaving devices. However, erroneous outputs, e.g., nuisance alerts, that may reduce crew attention and potentially cause serious consequences, must be addressed in the design.

Hence, human factors research is another future direction. This would include conducting testing that will assess the cyber aid usability and functionality in lab and training facility, and operational environments; evaluating crew threat situation awareness and response to alerts; and investigating subjective criteria for nuisance alerts. Human factors testing will give valuable feedback for cyber aid development, including tuning of algorithms and ATC crew display interface development.

B. Challenges with Using Kalman Filter Based Approach

A few challenges were observed during the implementation of the Kalman Filter based approach for ADS-B attack detection. In contrast to other studies, we chose the direction of examining entire flight segments which include all phases of flight. This direction posed a sizable challenge because it introduced a sizable amount of prediction errors which are indicated by the large spikes in Figure #. For example, when the aircraft is in the departure and climb phases of flight the rapid increase in altitude over a short period of time creates a noticeable amount of error realized by the KF. The same is true for the decent and approach phases of flight.

Another challenge with using the KF based approach on ADS-B data is the KF is not able to observe each message element all at the same time. For instance, we focused primarily on altitude, speed, and direction where a 1-D KF was applied to each parameter. We assumed an attacker would modify these specific ADS-B message parameters for our attack scenario; however, it would add complexity if an attacker modified every ADS-B message parameter. This would create a substantial challenge to implement the KF because one would have to use separate KFs as we did and develop a way to cross-check the results of each filter to derive a meaningful result.

Furthermore, a challenge was automating the detection of the KF so that it could be more user friendly for ATCOs to adopt and utilize. The KF proved to be an advantageous method to detect ADS-B anomalies, but the implementation of the KF only allows a user to observe the results in the form of a graph or measurements. We discuss in the next section the use of an additional machine learning neural network to work alongside the KF to better automate the detection of anomalies and to provide the user with clear actionable intelligence.

C. Future Research Directions

The proposed ADS-B threat detection capability and usability can be extended with the addition of neural network. Currently, the proposed system has two primary limitations: automation of threat detection and a user interface. The addition of a feedforward or other neural network would help eliminate these shortcomings of the current approach.

The feedforward neural network would be utilized to automate the discrimination task by training and testing the network on the estimation error covariances output from the Kalman Filters. In addition, we would want to develop a user interface so ATCOs can easily interact with the ML-based cyber aid system without increasing their workload.

An additional research direction would be to integrate more of the ADS-B message parameters, starting with latitude and longitude. We chose to exclude these two parameters from the current proposed system for a couple of reasons. The first being that GPS coordinates are measured by multiple satellites and there are uncontrollable factors which exist that can cause inaccurate measurements or no measurement at all. This is not to say GPS is inaccurate, but it does not rely on a singular system to provide the measurements. We decided to focus on altitude, speed, and direction because they are physical measurements that can be measured by instruments in the aircraft and be cross verified by other means. Further, these parameters are included in an aircraft's flight plan which can also be used to verify measurements.

Another reason latitude and longitude were not considered in this implementation was that the geometry becomes more complicated to understand. The GPS coordinates would need to be translated into 2-D position measures first to use them with the Kalman filter. Further, the use of GPS coordinates adds more error into the Kalman Filter because the measurements are noisy. However, we would like to explore the possibility of adding these positional parameters into the implementation to provide increased model accuracy and detection capability.

IX. CONCLUSIONS

In this paper, we proposed a crew-based approach to securing future ATC systems. Compared to the current state, we considered the risk from cyber-physical threats and the need to incorporate cyber resiliency into ATC systems. Both considerations warrant the consideration of crew and their contributions to early attack detection and incident response. We explored the established role of crew in operational environments and presented ways in which cyber security can be introduced into the education and training of ATCOs.

ADS-B was focused due to its central role in the NextGen and other future ATC systems across the world. We discussed cyber-physical attack scenarios on ATC system due to vulnerabilities of the ADS-B system architecture. We illustrated how incidents may appear to the ATCOs on their screens and engage them. We identified some high risk threats that may impact crew situation awareness and workload, and proposed the need for a cyber aid for crew in these scenarios.

We used an existing simulator, called OpenScope, that incorporates ADS-B cyberattack scenarios and serves as an experiment platform to raise ATC cyber threat awareness. This enabled us to educate the community about ATC cyber security challenges in our capture-the-flag competitions. However, for ATCO cyber education and readiness assessment, we identified the need for an experiment platform and setting that resembles practical demands and constraints of ATC systems. We also provided some experiment goals and outcomes but noted that our experiment and results will be reported as future work.

We explored the use of a machine learning model to detect spoofed ADS-B data using a Kalman Filter, which provides ATCOs with a tool to enhance cybersecurity situational awareness. The Kalman filter is an unconventional choice for a machine learning algorithm but nevertheless operates similar to a traditional learning algorithm all the same. The Kalman filter uses previous data measurements to predict the next point in time and it uses the covariance as its memory. One of the biggest advantages of the Kalman filter approach is the algorithm does not require vast amounts of data to draw complete conclusions.

We collected real-world ADS-B messages datasets and generated realistic pseudo-random attack data to demonstrate the ML-based cyber aid's ability to discriminate between normal and anomalous. Our performance evaluation results indicate that the ML-based cyber aid application detects spoofed ADS-B messages with a high probability of detection and a low probability of false alarm.

We have provided a means of understanding the results of the Kalman filter through the creation of a threat score, thresholds, and a trustworthiness scale. These three components can be used by ATCOs to draw rapid conclusions about the data thus enhancing cybersecurity situational awareness. Further, the trustworthiness scale can assist in the type of incident response required to ensure uninterrupted ATC operations. Overall, the Kalman Filter based approach proved to be an effective and advantageous solution to detect ADS-B anomalies and an applicable cyber aid for ATCOs.

ACKNOWLEDGMENT

The authors thank Dr. Farid Dowla, from the Lawrence Livermore National Lab, for his guidance on the machine learning model and algorithm development work presented in this paper. The authors also thank Prof. Kyle Wilkerson from ERAU-Prescott for providing us access to and helping with running scenarios in the ATC lab on campus. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of the DoD.

REFERENCES

- [1] K. Sampigethaya and R. Poovendran, "Aviation Cyber-Physical Systems: Foundations for Future Aircraft and Air Transport," in *Proceedings of the IEEE*, vol. 101, no. 8, pp. 1834-1855, Aug. 2013, doi: 10.1109/JPROC.2012.2235131.
- [2] Hatton, L. and Rutkowski, A. (2019). Lessons Must Be Learned-But Are They? *IEEE Software*, 36(4), 91-95.
- [3] J. Buch, R. Geister, L., Canzian, G., Gamba, & O. Pozzobon, "What the Hack Happened to the Flight Deck: Analyzing the Impact of Cyber Attacks on Commercial Flight Crews," In *AIAA Scitech 2019 Forum* (p. 0060).
- [4] World Economic Forum, "Global Risks Report 2022," <https://www.weforum.org/reports/global-risks-report-2022>.
- [5] Worldwide, Capgemini. "Reinventing cybersecurity with artificial intelligence: the new frontier in digital security." (2020). Available Online: <https://www.capgemini.com/>
- [6] T. E. Lyngset, "Securing critical ATC systems and protecting valuable data from Cyber Threats," World-class recording solution for ATC and Bank & Finance, 16-Oct-2020. [Online]. Available: <https://www.guardrec.com/blog/securing-critical-atc-systems-and-protecting-valuable-data-from-cyber-threats>. [Accessed: 23-Feb-2023].

- [7] M. Rupp, P. Smirnov, and U. Scholten, "Cybersecurity & Air Traffic Control : Revisiting techniques used in cyber-attacks and cyber-defense," *Short & Medium Range Radars for Training and Civil Aviation*, 09-Mar-2022. [Online]. Available: <https://www.skyradar.com/blog/cybersecurity-and-air-traffic-control-a-general-reminder-of-the-techniques-used-in-cyber-attacks-and-cybersecurity>. [Accessed: 23-Feb-2023].
- [8] A. Tamimi, A. Hahn, and S. Roy, "Cyber threat impact analysis to air traffic flows through Dynamic Queue Networks," *ACM Transactions on Cyber-Physical Systems*, vol. 4, no. 3, pp. 1–22, 2020.
- [9] G. Lykou, G. Iakovakis, and D. Gritzalis, "Aviation cybersecurity and cyber-resilience: Assessing risk in air traffic management," *Advanced Sciences and Technologies for Security Applications*, pp. 245–260, Jan. 2019.
- [10] "FAA Needs a More Comprehensive Approach to Address Cybersecurity As Agency Transitions to NextGen ," United States Government Accountability Office, Apr-2015. [Online]. Available: <https://www.gao.gov/assets/gao-15-370.pdf>. [Accessed: 23-Feb-2023].
- [11] "FAA Should Fully Implement Key Practices to Strengthen Its Oversight of Avionics Risks," United States Government Accountability Office, Apr-2015. [Online]. Available: <https://www.gao.gov/assets/gao-15-370.pdf>. [Accessed: 23-Feb-2023].
- [12] K. Sampigethaya, R. Poovendran and L. Bushnell, "Secure Operation, Control, and Maintenance of Future E-Enabled Airplanes," in *Proceedings of the IEEE*, vol. 96, no. 12, pp. 1992-2007, Dec. 2008, doi: 10.1109/JPROC.2008.2006123.
- [13] S. Pullen, G. Gao, C. Tedeschi, & J. Warburton, "The impact of uninformed RF interference on GBAS and potential mitigations," in *Proceedings of the 2012 International Technical Meeting of The Institute of Navigation* (pp. 780-789).
- [14] "B777 in Autoland Mode Left Runway When Another Aircraft Interfered With the Localiser Signal (9V-SWQ, SIA)," <https://aerossurance.com/safety-management/sia-b777-autoland-localiser-re/>.
- [15] ATO Cybersecurity Group, "ATO Cybersecurity Group | Federal Aviation Administration, 31-Aug-2022. [Online]. Available: https://www.faa.gov/air_traffic/technology/cas/acg. [Accessed: 24-Feb-2023].
- [16] J. H. Williams and T. L. Signore, "National Airspace System Security Cyber Architecture - Mitre Corporation," MITRE Publications. [Online]. Available: https://www.mitre.org/sites/default/files/publications/10_4169.pdf. [Accessed: 23-Feb-2023].
- [17] M. Moon, "FAA's NOTAM computer outage affected military flights," *Engadget*, 14-Jan-2023. [Online]. Available: <https://www.engadget.com/faa-notam-computer-outage-affected-even-military-flights-155514704.html>. [Accessed: 23-Feb-2023].
- [18] "FAA Notam Statement," *FAA NOTAM Statement | Federal Aviation Administration*, 19-Jan-2023. [Online]. Available: <https://www.faa.gov/newsroom/faa-notam-statement>. [Accessed: 23-Feb-2023].
- [19] D. Goward, "What happened to GPS in Denver?," *GPS World*. [Online]. Available: <https://www.gpsworld.com/what-happened-to-gps-in-denver/>. [Accessed 23-Feb-2023].
- [20] D. Goodin, "GPS interference caused the FAA to reroute Texas air traffic. Experts stumped," *Ars Technica*. [Online]. Available: <https://arstechnica.com/information-technology/2022/10/cause-is-unknown-for-mysterious-gps-outage-that-rerouted-texas-air-traffic/>. [Accessed 23-Feb. 2023]
- [21] Faa.gov, "Runway safety statistics." [Online]. Available: http://www.faa.gov/airports/runway_safety/statistics/
- [22] NASA, "Evaluating the Effectiveness of Schedule Changes for Air Traffic Service (ATS) Providers: Controller Alertness and Fatigue Monitoring Study." [Online]. Available: http://www.faa.gov/data_research/research/media/nasa_controller_fatigue_assessment_report.pdf
- [23] Performance-based operations Aviation Rulemaking Committee, "Operational use of flight management systems." [Online]. Available:

- http://www.faa.gov/about/office_org/headquarters_offices/avs/offices/af/safs400/parc/parc_reco/media/2013/130908_parc_flt dawg_final_report_recommendations.pdf
- [24] Airline Pilots Association (ALPA), "Aircraft Cyber Security: The Pilot's Perspective," Technical report, Air Line Pilots Association Int'l, Washington, 2017. [Online]. https://www.rtca.org/sites/default/files/symposium_2017_cybersecurity_white_paper_digital.pdf
- [25] P. Gontar, H. Homans, et. al., "Are pilots prepared for a cyber-attack? A human factors approach to the experimental evaluation of pilots' behavior," *Journal of Air Transport Management*, vol. 69, pp. 26-37, ISSN 0969-6997, 2018.
- [26] Nguyen, D., Shelton, J. W., & Mitchell, T. M. "System and method for evaluating cyber- attacks on aircraft." (2017) *U.S. Patent No. 9,836,990*. Washington, DC: U.S. Patent and Trademark Office.
- [27] Raytheon's Cyber Anomaly Detection System (CADS), Intrusion Detection System for the 1553 Bus. [Online]. <https://www.raytheonintelligenceandspace.com/what-we-do/cybersecurity/data-resiliency/cads>.
- [28] E. Nystad, J. E. Simensen and C. Rasputnig, "Investigating operative cybersecurity awareness in air traffic control," 2021 14th International Conference on Security of Information and Networks (SIN), Edinburgh, United Kingdom, 2021, pp. 1-8, doi: 10.1109/SIN54109.2021.9699158.
- [29] J. Lopez, D. Otto, "Analysis of cybersecurity content in the air traffic collegiate training initiative (AT-CTI) program." *Two Cultures*, 1(1), pp. 1-17, 2012.
- [30] FAA, "The air traffic controller workforce plan: 2022-2031." [Online]. Available: <https://www.faa.gov/sites/faa.gov/files/2022-06/2022-afn-cwp.pdf>.
- [31] OpenScope Air Traffic Management Simulator. [Online] Available: <https://www.openscope.co>. (Accessed on 27/3/2020).
- [32] A. Blåberg, G. Lindahl, A. Gurtov, & B. Josefsson, "Simulating ADS-B attacks in air traffic management," in 2020 AIAA/IEEE 39th Digital Avionics Systems Conference (DASC) (pp. 1-10). IEEE.
- [33] J. Wang, Y. Zou, and J. Ding, "ADS-B spoofing attack detection method based on LSTM," *EURASIP Journal on Wireless Communications and Networking*, vol. 2020, no. 1, 2020.
- [34] X. Ying, J. Mazer, G. Bernieri, M. Conti, L. Bushnell, and R. Poovendran, "Detecting ADS-B spoofing attacks using Deep Neural Networks," 2019 *IEEE Conference on Communications and Network Security (CNS)*, Apr. 2019.
- [35] S. Khan, J. Thorn, A. Wahlgren, and A. Gurtov, "Intrusion detection in automatic dependent surveillance-broadcast (ADS-B) with machine learning," 2021 *IEEE/AIAA 40th Digital Avionics Systems Conference (DASC)*, 2021.
- [36] E. Habler and A. Shabtai, "Using LSTM encoder-decoder algorithm for detecting anomalous ADS-B messages," *Computers & Security*, vol. 78, pp. 155-173, Jul. 2018.